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1. IPC model that does not exchange of data between processes is _____

Hide Results 41/43 Students Answered

A	Message Passing	0%
✓ B	Signal	98%
C	Shared Memory	0%
D	A or B	2%
E	B or C	0%

3. IPC model that allow processes to exchange data but the synchronization is the responsibility of programmers is _____

Hide Results 41/43 Students Answered

A	Message Passing	7%
B	Signal	2%
✓ C	Shared Memory	88%
D	A or B	0%

5. A signal can be generated from _____.

Hide Results 41/43 Students Answered

A	Hardware	0%
B	Processes	2%
C	User	2%
D	A or B	2%
✓ E	A or B or C	93%

7. To catch a signal, a programmer writes a function that will be called when the signal is sent to the process.

Hide Results 41/43 Students Answered

✓	True	95%
	False	5%

9. The signal that will be sent to the process when a user press Ctrl-c is _____

Hide Results 41/43 Students Answered

A	SIGFPE	0%
✓ B	SIGINT	98%
C	SIGTERM	0%
D	SIGQUIT	0%
E	SIGKILL	2%

[Back to Results Table](#)

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2. IPC model that allow processes to exchange data and the synchronization is managed by OS is _____

Hide Results 41/43 Students Answered

✓ A	Message Passing	85%
B	Signal	0%
C	Shared Memory	12%
D	A or B	2%

4. Message passing mechanism that is provided by the UNIX operating system is _____.

Hide Results 41/43 Students Answered

✓ A	pipe	73%
B	signal	5%
C	shared memory	0%
D	A or B	22%
E	B or C	0%

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6. In general, when a process receives a signal it can _____.

Hide Results 41/43 Students Answered

A	catch that signal	5%
B	mask that signal	2%
C	use OS default for that signal	0%
D	A or B	2%
✓ E	A or B or C	90%

8. There are signals that cannot be caught or disabled.

Show Results 41/43 Students Answered

True	95%
False	5%

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10. System call signal() is used to send a signal from process to process.

Hide Results 41/43 Students Answered

True	5%
✓ False	95%

11. A signal handler function in a UNIX like operating system must accept one parameter of type int which is the signal that is sending to the process.

Hide Results

41/43 Students Answered

True 93%

False 7%

13. To disable the incoming signal in a signal handler, the macro that you will pass as the second parameter to the signal() function is _____

Hide Results

41/43 Students Answered

A SIG_DIS 10%

B SIG_FPE 2%

☒ C SIG_IGN 88%

D SIG_INT 0%

E SIG_QUIT 0%

15. System call raise() is used to send a signal from one process to another.

Hide Results

41/43 Students Answered

True

False

```

1 #include <stdio.h>
2 #include <signal.h>
3 void SIGHandler(int);
4 void main(void)
5 {
6     signal(SIGINT, SIGHandler);
7     signal(SIGQUIT, SIGHandler);
8     signal(SIGKILL, SIGHandler);
9     while (1); /* Loop forever */
10 }
11 void SIGHandler(int sig)
12 {
13     char c;
14     signal(SIGINT, SIG_IGN);
15     signal(SIGQUIT, SIG_IGN);
16     signal(SIGKILL, SIG_IGN);
17     switch(sig) {
18     case SIGINT:
19         printf("Ha! Ha! You cannot interrupt me\n");
20         break;
21     case SIGQUIT:
22         printf("Ha! Ha! you cannot quit me\n");
23         break;
24     case SIGKILL:
25         printf("Ha! Ha! you cannot kill me\n");
26         break;
27     }
28     signal(SIGINT, SIGHandler);
29     signal(SIGQUIT, SIGHandler);
30     signal(SIGKILL, SIGHandler);
31 }

```

18. From the C program in the picture;

Assuming that after running the program, the process has 1234 as its process ID. If a user enters the following command kill -9 1234, what is the result of the program?

Hide Results

41/43 Students Answered

[Back to Results Table](#)

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>

12. You must disable all signals in a signal handler function.

Hide Results

41/43 Students Answered

True 0%

☒ False 100%

14. A process can handle only one signal.

Hide Results

41/43 Students Answered

True 2%

☒ False 98%

[Back to Results Table](#)

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>

16. System call kill() is used only to send the signal SIGKILL from one process to another.

Hide Results

41/43 Students Answered

True 20%

☒ False 80%

17. From the C program in the picture;

If a user presses Ctr-c after running the program, what is the result of the program?

Hide Results

41/43 Students Answered

☒ A Ha! Ha! You cannot interrupt me 90%

B Ha! Ha! You cannot quit me 0%

19. From the C program in the picture;

Assuming that after running the program, the process has 1234 as its process ID. If a user enters the following command kill -INT 1234, what is the result of the program?

Hide Results

41/43 Students Answered

☒ A Ha! Ha! You cannot interrupt me 90%

B Ha! Ha! You cannot quit me 2%

C Ha! Ha! You cannot kill me 24%

☒ D The program is killed. 66%

E A runtime error message is sent. 7%

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1. The message-passing function that is used to send data must be named send()

Hide Results

41/43 Students Answered

True	22%
False	78%

< Back to Results Table

3. What is (are) considered the synchronous passing?

Hide Results

41/43 Students Answered

A	a sender waits until the receiver gets the message.	0%
B	sender sends the message and continue.	2%
C	a receiver waits until a message is available.	0%
D	a receiver does not wait for a message.	0%
E	A and B	0%
F	A and C	98%
G	B and C	0%

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5. The parameter that will be passed to the function to create a pipe is _____.

Hide Results

41/43 Students Answered

A	A string that is a filename that will be used as a pipe.	5%
B	An array of integer that has 2 elements.	90%
C	An integer key that is used to specify a unique pipe.	2%
D	An integer variable that is used to receive an address of a pipe.	2%

< Back to Results Table

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7. After creating a pipe, we get file descriptors and use them to work with the pipe.

Hide Results

41/43 Students Answered

True	95%
False	5%

9. If a parent needs to communicate with its child, it will fork the child process, create a pipe, and use that pipe to communicate with the child.

Hide Results

41/43 Students Answered

True	80%
False	20%

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< 2 of 24 >

2. Message passing can be only unidirectional.

Hide Results

41/43 Students Answered

True	2%
False	98%

4. The function to create a pipe is _____.

Hide Results

41/43 Students Answered

A	create_pipe()	0%
B	createPipe()	0%
C	pipe_create()	0%
D	pipeCreate()	0%
E	pipe()	100%

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6. We use the concept of writing and reading a file when working with pipe.

Hide Results

41/43 Students Answered

True	100%
False	0%

8. If pfd is a variable, we pass it as the parameter when creating a pipe. Assume that the data that we need to write to the pipe is stored in the array named data, and the length of the data is stored in the variable called len. We use _____ statement to write to the pipe. (Choose all that apply)

Hide Results

41/43 Students Answered

A	send(pfd[0], data, length);	0%
B	send(pfd[1], data, length);	0%
C	write(pfd[0], data, length);	5%
D	write(pfd[1], data, length);	95%

Library #define SIZE 1024

```

int main() {
    int pfd[2];
    pid_t pid;
    char data[SIZE];

    //a create pipe
    //b fork a child process
    if (pid == 0) {
        //c close the read end
        strcpy(data, "hello ");
        //d write to the parent
        //e close the write end
        //f exit successfully
    }
    else {
        //g close the write end
        //h read from pipe and store data into data
        printf("parent read %s\n", data);
        //i wait for child
        //j close the read end
    }
}

```

11. Assuming that we need to write a program to send data from the child process to the parent

The statement at //b is

41/43 Students Answered

- A `close(pfd[0]);`
- B `close(pfd[1]);`
- C `pipe(pfd);`
- D `pid = fork();`
- E `fork(pid);`

13. Assuming that we need to write a program to send data from the child process to the parent

The statement at //d is

41/43 Students Answered

- A `send(pfd[0], data, strlen(buf) + 1);` 0%
- B `send(pfd[1], data, strlen(buf) + 1);` 0%
- C `write(pfd[0], data, strlen(buf) + 1);` 10%
- D `write(pfd[1], data, strlen(buf) + 1);` 90%

15. Assuming that we need to write a program to send data from the child process to the parent

The statement at //f is

41/43 Students Answered

- A `close(pfd[0]);` 2%
- B `close(pfd[1]);` 0%
- C `exit(0);` 95%
- D `wait(NULL)` 2%
- E `fork();` 0%

17. Assuming that we need to write a program to send data from the child process to the parent

The statement at //h is

41/43 Students Answered

- A `read(pfd[0], data, SIZE);` 90%
- B `read(pfd[1], data, SIZE);` 5%
- C `receive(pfd[0], data, SIZE);` 5%
- D `receive(pfd[1], data, SIZE);` 0%
- E `receive(pfd, data, SIZE);` 0%

12. Assuming that we need to write a program to send data from the child process to the parent

The statement at //c is

41/43 Students Answered

- A `close(pfd[0]);` 93%
- B `close(pfd[1]);` 5%

14. Assuming that we need to write a program to send data from the child process to the parent

The statement at //e is

41/43 Students Answered

- A `close(pfd[0]);` 7%
- B `close(pfd[1]);` 88%
- C `exit(0);` 5%
- D `wait(NULL)` 0%
- E `fork();` 0%

16. Assuming that we need to write a program to send data from the child process to the parent

The statement at //g is

41/43 Students Answered

- A `close(pfd[0]);` 5%
- B `close(pfd[1]);` 93%
- C `exit(0);` 2%
- D `wait(NULL)` 0%
- E `fork();` 0%

18. Assuming that we need to write a program to send data from the child process to the parent

The statement at //i is

41/43 Students Answered

- A `close(pfd[0]);` 0%
- B `close(pfd[1]);` 2%
- C `exit(0);` 2%
- D `wait(NULL)` 95%
- E `fork();` 0%

②
6.9.2024
Message Passing

20. The file descriptor that represents standard input is ____.

[Hide Results](#)

41/43 Students Answered

✓ A	0	98%
B	1	2%
C	2	0%

19. Assuming that we need to write a program to send data from the child process to the parent

The statement at //j is

③
G.Q.24
Message Passing

[Hide Results](#)

41/43 Students Answered

✓ A	close(pfd[0]);	88%
B	close(pfd[1]);	10%

22. The default file descriptor that the `ls` command uses to display data is ____.[Hide Results](#)

41/43 Students Answered

A	0	2%
✓ B	1	88%
C	2	0%
D	3	5%
E	4	5%

21. The file descriptor that represents standard output is ____.

[Hide Results](#)

41/43 Students Answered

A	0	5%
✓ B	1	95%
C	2	0%

23. If we need to redirect the standard output to the write-end of the pipe, the statement that we will use is _____. (Choose all that apply)

[Hide Results](#)

41/43 Students Answered

A	dup2(pfd[0],0);	0%
B	dup2(pfd[0],1);	5%
C	dup2(pfd[1],0);	7%
✓ D	dup2(pfd[1],1);	88%

24. If we need to redirect the standard input to the read-end of the pipe, the statement that we will use is _____. (Choose all that apply)

[Hide Results](#)

41/43 Students Answered

✓ A	dup2(pfd[0],0);	93%
B	dup2(pfd[0],1);	2%
C	dup2(pfd[1],0);	5%

10. Assuming that we need to write a program to send data from the child process to the parent

The statement at //a is

[Hide Results](#)

41/43 Students Answered

A	close(pfd[0]);	5%
B	close(pfd[1]);	0%
✓ C	pipe(pfd);	93%
D	pipe(pfd[0]);	2%
E	pipe(pfd[1]);	0%

1. The system call that a process has to use to create a shared memory is _____.

Hide Results

40/43 Students Answered

A	shmat()	0%
B	shmctl()	3%
C	shmdt()	0%
✓ D	shmget()	98%
E	IPC_PRIVATE	0%

3. The first parameter of the system call to create a shared memory is _____.

Hide Results

40/43 Students Answered

✓ A	shared memory key	95%
B	shared memory id	5%

5. If a unique **key is needed**, _____ will be used as the first parameter of the system call(function) we use to create a shared memory.

Hide Results

40/43 Students Answered

A	IPC_CREAT	3%
B	IPC_FLAG	0%
✓ C	IPC_PRIVATE	83%
D	IPC_CREAT 0666	3%
E	A or D	13%

7. The limitation of using IPC_PRIVATE as a shared memory key is that it can be used only between a parent process and its children processes, i.e., a child process of its children processes cannot use the shared memory created by the parent.

Hide Results

40/43 Students Answered

	True	53%
✓	False	48%

9. If we need a shared memory to store an array of doubles with 5 elements, the parameter of the shmget() function will be specified as _____.

Hide Results

40/43 Students Answered

A	5	0%
B	5 * double	0%
✓ C	5 * sizeof(double)	100%
D	double[5]	0%
E	B or C	0%

2. The system call that a process that needs to use the shared memory created by another process is _____.

Hide Results

40/43 Students Answered

A	shmat()	88%
B	shmctl()	3%
C	shmdt()	3%
✓ D	shmget()	8%
E	IPC_PRIVATE	0%

4. The function we can use to create the parameter's value of the system call to create a shared memory is _____.

Hide Results

40/43 Students Answered

A	shmat()	0%
B	shmdt()	0%
C	shmctl()	3%
✓ D	ftok()	95%
E	keygen()	3%

6. The function to create a shared memory returns a (an) _____.

Hide Results

40/43 Students Answered

A	shared memory key	0%
✓ B	shared memory id	98%
C	shared memory address	3%
D	shared memory control	0%
E	shared memory detach	0%

8. The second parameter of the function **shmget()** is the specification of _____.

Hide Results

40/43 Students Answered

A	permission to use the shared memory e.g. read only or read and write	5%
✓ B	size of the shared memory	90%
C	number of processes that can use the shared memory	0%

10. If we need to specify permission of a file as follows **rw-r--** the octal number that we need to pass as a parameter to the chmod command is _____

Hide Results

40/43 Students Answered

A	0666	8%
B	0644	0%
C	0664	5%
D	0674	0%
✓ E	0764	88%

11. The function to attach the process address space to a shared memory's address is _____.

Hide Results

40/43 Students Answered

✓ A	shmat()	100%
B	shmdt()	0%
C	shmctl()	0%
D	shm_ptr()	0%
E	shmget()	0%

13. The first parameter of the function `shmctl()` is the value returned by the _____ function.

Hide Results

40/43 Students Answered

A	shmdt()	0%
✓ B	shmget()	100%
C	shmctl()	0%
D	malloc	0%

15. The function to remove a shared memory is _____.

Hide Results

40/43 Students Answered

A	shmat()	0%
B	shmdt()	13%
C	shmget()	5%
✓ D	shmctl()	83%
E	shm_ptr()	0%

17. After a parent process creates and attaches itself to a shared memory and then forks its child, the child process can access that shared memory.

Hide Results

40/43 Students Answered

✓ True	95%
False	5%

19. Which statement(s) can be used by a process that creates a shared memory and wants to allow any processes that use the same key to access the shared memory?

Hide Results

40/43 Students Answered

A	<code>shmid = shmget(IPC_PRIVATE, sizeof(struct data), IPC_CREAT 0666);</code>	5%
✓ B	<code>shmid = shmget(ftok(".", "a"), sizeof(struct data), IPC_CREAT 0666);</code>	95%
C	<code>shmid = shmget(IPC_PRIVATE, sizeof(struct data), 0666);</code>	0%

12. The variable's data type, which stores the address of a shared memory, must be a pointer to an integer.

②

13-9-24

Shared Memo

Hide Results

40/43 Students Answered

True	18%
False	83%

14. If a shared memory is created to store data of type `double`, the variable to access shared memory needs to be declared as _____.

Hide Results

40/43 Students Answered

A	<code>int shm_ptr;</code>	0%
B	<code>int *shm_ptr;</code>	0%
C	<code>double shm_ptr;</code>	0%
✓ D	<code>double *shm_ptr;</code>	100%

16. The command to pass as a parameter to remove a shared memory by using the function in the previous question is _____.

Hide Results

40/43 Students Answered

A	<code>IPC_DEL</code>	3%
B	<code>IPC_REM</code>	0%
✓ C	<code>IPC_RMID</code>	95%
D	<code>IPC_CTL</code>	0%
E	<code>IPC_DISP</code>	3%

run lintaloksum

18. Using `ftok()` to create a shared memory key can prevent other processes from making the same shared memory key as our process.

Hide Results

40/43 Students Answered

True	13%
✓ False	88%

20. Which statement(s) can be used by a process that creates a shared memory that will be used only by processes in the same family (parent/child/sub-child)?

Hide Results

40/43 Students Answered

✓ A	<code>shmid = shmget(IPC_PRIVATE, sizeof(struct data), IPC_CREAT 0666);</code>	90%
B	<code>shmid = shmget(ftok(".", "a"), sizeof(struct data), IPC_CREAT 0666);</code>	3%
C	<code>shmid = shmget(IPC_PRIVATE, sizeof(struct data), 0666);</code>	8%

21. If a shared memory is created using the following statement;

```
shmid = shmget(ftok(".", "a"), sizeof(struct data),  
IPC_CREAT | 0666);
```

The variable that will access the shared memory must be declared _____.

Hide Results

40/43 Students Answered

- | | | |
|-----|-----------------------|------|
| A | int *shm_ptr; | 0% |
| B | int shm_ptr; | 0% |
| C | struct data shm_ptr; | 0% |
| ✓ D | struct data *shm_ptr; | 100% |
| E | double *shm_ptr; | 0% |

23. If we need to detach the process from a shared memory, we will use the statement _____.

Hide Results

40/43 Students Answered

- | | | |
|-----|----------------------------------|-----|
| A | shmat((void *) shm_ptr); | 0% |
| ✓ B | shmdt((void *) shm_ptr); | 85% |
| C | shmat(shmid); | 0% |
| D | shmdt(shmid); | 15% |
| E | shmctl(shm_ptr, IPC_RMID, NULL); | 0% |

22. The statement that allows the variable declared in the previous question to access the shared memory is _____.

Hide Results

40/43 Students Answered

- | | | |
|-----|--|-----|
| A | shm_ptr = (int *) shmat(shmid, NULL, 0); | 5% |
| B | shm_ptr = (int *) shmdt(shmid, NULL, 0); | 0% |
| ✓ C | shm_ptr = (struct data *) shmat(shmid, NULL, 0); | 93% |
| D | shm_ptr = (struct data *) shmdt(shmid, NULL, 0); | 3% |
| E | shm_ptr = (double *) shmat(shmid, NULL, 0); | 0% |

24. If we need to remove a shared memory, we will use the statement _____.

Hide Results

40/43 Students Answered

- | | | |
|-----|----------------------------------|-----|
| A | shmctl(shm_ptr, IPC_RMID, NULL); | 3% |
| ✓ B | shmctl(shmid, IPC_RMID, NULL); | 93% |
| C | shmrm(shm_ptr, IPC_RMID, NULL); | 5% |
| D | shmrm(shmid, IPC_RMID, NULL); | 0% |

3

13.9.24

Shared Memo

1. Synchronization method that works by putting a process to wait inside a loop is called busy waiting

Hide Results 41/43 Students Answered

✓ True	98%
False	2%

3. Semaphore that is used to solve critical section problem is counting semaphore

Hide Results 41/43 Students Answered

True	10%
False	90%

5. For using semaphore to solve a critical section, consider the following pseudo code;

Semaphore S;

//1

Critical Section

//2

S must be initialized with this value.

Hide Results 41/43 Students Answered

A 0	5%
✓ B 1	95%
C 2	0%
D 3	0%

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7. For using a semaphore to solve a critical section, consider the following pseudo code;

Semaphore S;

//1

Critical Section

//2

After the semaphore S is initialized 1, what should be the code at 2?

2. Semaphore is an OS synchronization that does not require busy waiting

Hide Results 41/43 Students Answered

✓ True	95%
False	5%

4. Binary semaphore can be called ____

Hide Results 41/43 Students Answered

A Counting Semaphore	2%
✓ B Mutex lock	98%
C wait	0%
D signal	0%
E synchronization	0%

consider the following pseudo code,

Semaphore S;

//1

Critical Section

//2

After the semaphore S is initialized 1, what should be the code at 1?

Hide Results 41/43 Students Answered

✓ A wait(S);	100%
B signal(S)	0%
C sleep(S);	0%
D join(S)	0%
E exit(S)	0%

8. The function of PThread Semaphore that is used as the semaphore wait() is ____

Hide Results 41/43 Students Answered

A sem_init()	2%
B sem_post()	0%
✓ C sem_wait()	98%
D sem_signal()	0%
E sem_join()	0%

same ques

9. The function of PThread Semaphore that is used as the semaphore signal() is _____

Hide Results

41/43 Students Answered

A	sem_init()	2%
✓ B	sem_post()	93%
C	sem_wait()	0%
D	sem_signal()	5%
E	sem_join()	0%

11. From the C program in the picture
The out out is _____

Hide Results

41/43 Students Answered

A	3	7%
✓ B	4	24%
C	5	69%

```

1 #include <pthread.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <unistd.h>
5 #include <semaphore.h>
6 int count = 5;
7 sem_t mutex;
8 void *increment(void *param);
9 void *decrement(void *param);
10 int main() {
11     int i = 0;
12     pthread_t t1, t2;
13     pthread_t tid[5];
14     pthread_attr_t attr;
15     pthread_attr_t attr;
16     sem_init(&mutex, 0, 1);
17     for(int i = 0; i < 5; i++) {
18         if ((i%2)) {
19             pthread_create(&tid[i], &attr, increment, NULL);
20         }
21         else {
22             pthread_create(&tid[i], &attr, decrement, NULL);
23         }
24     }
25     for(int i = 0; i < 5; i++) {
26         pthread_join(tid[i], NULL);
27     }
28     printf("count = %d\n", count);
29     sem_destroy(&mutex);
30 }
31 void *increment (void *param) {
32     int temp;
33     sem_wait(&mutex);
34     temp = count;
35     sleep(rand() % 5);
36     temp++;
37     count = temp;
38     sem_post(&mutex);
39     pthread_exit(0);
40 }
41 void *decrement (void *param) {
42     int temp;
43     sem_wait(&mutex);
44     temp = count;
45     sleep(rand() % 5);
46     temp--;
47     count = temp;
48     sem_post(&mutex);
49     pthread_exit(0);
50 }

```

10. The function of PThread Semaphore that is used to initialize a semaphore is _____

Hide Results

41/43 Students Answered

✓ A	sem_init()	100%
B	sem_post()	0%
C	sem_wait()	0%
D	sem_signal()	0%
E	sem_join()	0%

12. In our course, how many semaphore(s) is(are) used to solve bounded buffer problem?

Hide Results

41/43 Students Answered

A	1	0%
B	2	32%
✓ C	3	66%
D	4	2%
E	5	0%

13. In our course, how many counting semaphore(s) is (are) used to solve bounded buffer problem?

Hide Results

41/43 Students Answered

A	1	20%
✓ B	2	68%
C	3	12%
D	4	0%
E	5	0%

14. In our course, the semaphore that is used by the producer to check whether the buffer is full or not is _____

Hide Results

41/43 Students Answered

✓ A	empty	56%
B	full	44%
C	mutex	0%
D	wait	0%
E	signal	0%

20.9.2024
Semaphore

15. In our course, the semaphore that is used by the consumer to check whether a new data is available or not is _____

20.9.2024 (3)
Semaphore

Hide Results

41/43 Students Answered

A	empty	27%
✓ B	full	71%
C	mutex	2%
D	wait	0%
E	signal	0%

17. For reader-writer problem in this course, the semaphore that is used to prevent reader and writer to read/update data at the same time is _____

Hide Results

41/43 Students Answered

A	mutex	24%
✓ B	wrt	66%
C	readcount	7%
D	signal	2%
E	wait	0%

19. For reader-writer problem in this course, If readers start reading the data, writer has to wait until the last reader finished reading the data.

Hide Results

41/43 Students Answered

✓ True	98%
False	2%

21. For Dining philosopher problem in this course, each chopstick represents a _____ in a computer system.

Hide Results

41/43 Students Answered

A	process	0%
✓ B	resource	83%
C	semaphore	15%
D	synchronization	2%
E	monitor	0%

23. Using semaphore to dining philosopher problem as we did in this course, may cause _____

Hide Results

41/43 Students Answered

✓ A	deadlock	98%
B	overflow	0%
C	starvation	3%
D	race condition	0%
E	termination	0%

16. In our course, the semaphore that is used to ensure that the producer and the consumer will not update/access the buffer at the same time is _____

Hide Results

41/43 Students Answered

A	empty	5%
B	full	2%
✓ C	mutex	93%
D	wait	0%
E	signal	0%

18. For reader-writer problem in this course, the semaphore that is used to ensure that the creation of a new reader will be counted correctly is _____

Hide Results

41/43 Students Answered

✓ A	mutex	49%
B	wrt	5%
C	readcount	44%
D	signal	0%
E	wait	2%

20. For Dining philosopher problem in this course, each philosopher represents a _____ in a computer system.

Hide Results

41/43 Students Answered

✓ A	process	95%
B	resource	0%
C	semaphore	2%
D	synchronization	2%
E	monitor	0%

22. For Dining philosopher problem in this course, a semaphore is used to allocate a chopstick to a philosopher.

Hide Results

41/43 Students Answered

True	95%
✓ False	5%

27.9.2024 (Java Synchronization)

1. The main idea of monitor is to combine shared data and functions to work with shared data into one unit and OS will ensure that one process will be active within a monitor at a time.

Hide Results

39/43 Students Answered

True 100%

False 0%

2. The process that is active in a monitor, will be moved back to the entry queue when it needs to wait for some condition to occur.

Hide Results

39/43 Students Answered

True 23%

False 77%

3. Applying monitor to solve dining philosopher problem using the algorithm in this course may cause the problem of _____

Hide Results

39/43 Students Answered

A deadlock 18%

B overflow 3%

✓ C starvation 77%

D race condition 3%

E termination 0%

4. Java provides a mechanism that resembles the idea of monitor is _____

Hide Results

39/43 Students Answered

A semaphore 3%

B starvation 0%

✓ C synchronization 97%

D Thread 0%

E Runnable 0%

5. Java synchronization will be occurred at only the method level.

Hide Results

39/43 Students Answered

True 10%

✓ False 90%

7. What is the correct way to say that the method is a Java synchronized method?

Hide Results

39/43 Students Answered

A public void synchronized func() {} 5%

✓ B public synchronized void func() {} 92%

C public void func() implements synchronized {} 3%

D public void func() implements synchronized {} 0%

9. Calling the yield() method will cause the object to be unlocked to allow other threads to work with synchronized methods.

Hide Results

39/43 Students Answered

True 8%

✓ False 92%

6. Java synchronization is done by calling lock() and release() method.

Hide Results

39/43 Students Answered

True

✓ False

8. When a thread calls an object synchronized method, the thread will be put to the wait set if there is a thread that previously called the same or different synchronized method and does not finish executing that method.

Hide Results

39/43 Students Answered

True 92%

✓ False 8%

10. If wait() is called inside a Java synchronized method, thread will move to the entry set.

Hide Results

38/43 Students Answered

True 6%

✓ False 95%

11. A programmer can use method notify() to choose a specified process from the wait set.

Hide Results

39/43 Students Answered

True 36%

✓ False 64%

12. From Java program in the picture, what is the output?

Hide Results

38/43 Students Answered

- A 4 3%
- B 5 11%
- C 6 0%
- D 7 8%
- ✓ E 8 78%

14. From the following Java program in the picture, What is(are) the possible output?

Hide Results

38/43 Students Answered

- A increment increment decrement decrement 5
- B decrement decrement increment increment 5
- C decrement increment decrement increment 5
- D A or B
- ✓ E All of the above

16. From the BoundedBuffer class in the picture.
The statement at //B is ____

Hide Results

38/43 Students Answered

- A yield(); 3%
- B wait(); 0%
- C notify(); 5%
- ✓ D 0 87%
- E 5 5%

18. From the BoundedBuffer class in the picture.
The statement at //D is ____

Hide Results

38/43 Students Answered

- A yield(); 5%
- ✓ B wait(); 87%
- C notify(); 5%
- D 0 3%
- E 5 0%

13. From the following Java program in the picture.
What is(are) the possible output?

Hide Results

39/43 Students Answered

- A increment increment decrement decrement 5 15%
- B decrement decrement increment increment 5 8%
- C decrement increment decrement increment 5 5%
- ✓ D A or B 15%
- E All of the above. 56%

27.9
(Java Syn)

15. From the BoundedBuffer class in the picture.
The statement at //A is ____

Hide Results

39/43 Students Answered

- A yield(); 8%
- B wait(); 3%
- C notify(); 0%
- ✓ D 0 82%
- E 5 8%

17. From the BoundedBuffer class in the picture.
The statement at //C is ____

Hide Results

39/43 Students Answered

- A yield(); 0%
- B wait(); 3%
- C notify(); 3%
- ✓ D 0 79%
- E 5 16%

19. From the BoundedBuffer class in the picture.
The statement at //E is ____

Hide Results

38/43 Students Answered

- A yield(); 3%
- B wait(); 5%
- ✓ C notify(); 87%
- D 0 0%
- E 5 5%

20. From the BoundedBuffer class in the picture.
The statement at //F is ____

Hide Results

37/43 Students Answered

A yield();

✓ B wait();

C notify();

D 0

E 5

21. From the BoundedBuffer class in the picture.
The statement at //G is ____

Hide Results

38/43 Students Answered

A yield();

B wait();

✓ C notify();

D 0

E 5

```

BoundedBuffer.java
public class BoundedBuffer
{
    private static final int BUFFER_SIZE = 5;
    private int count; // number of items in the buffer
    private int in; // points to the next free position in the buffer
    private int out; // points to the next full position in the buffer
    private Object[] buffer;
    public BoundedBuffer()
    {
        // buffer is initially empty
        count = //A ;
        in = //B ;
        out = //C ;
        buffer = new Object[BUFFER_SIZE];
    }
    //producer calls this method
    public synchronized void insert(Object item) {
        while (count == BUFFER_SIZE) {
            try {
                //D;
            }
            catch (InterruptedException e) {
            }
        }
        buffer[in] = item;
        in = (in + 1) % BUFFER_SIZE;
        count++;
        if (count == BUFFER_SIZE)
            System.out.println("Producer Entered " + item +
                               " Buffer FULL");
        else
            System.out.println("Producer Entered " + item +
                               " Buffer Size = " + count);
        //E;
    }
}

```

```

// consumer calls this method
public synchronized Object remove() {
    Object item;
    while (count == 0) {
        try {
            //F;
        }
        catch (InterruptedException e) {
        }
    }
    item = buffer[out];
    out = (out + 1) % BUFFER_SIZE;
    count--;
    if (count == 0)
        System.out.println("Consumer Consumed " + item +
                             " Buffer EMPTY");
    else
        System.out.println("Consumer Consumed " + item +
                             " Buffer Size = " + count);
    //G;
    return item;
}

```

27.9 (Java Synchro)

27.9.2024 (Java Semaphore) mutex (init) = 1 , full (count) = 0 , empty (size of bounded buffer) = N

1. The method of the Java Semaphore class that functions similarly to wait() of Semaphore is

Hide Results

43/43 Students Answered

A	acquire()	86%
B	notify()	2%
C	release()	5%
D	signal()	0%
E	wait()	7%

2. The method of the Java Semaphore class that functions similarly to signal() of Semaphore is

Hide Results

43/43 Students Answered

A	acquire()	0%
B	notify()	9%
C	release()	86%
D	signal()	2%
E	wait()	2%

```
import java.util.concurrent.Semaphore;
public class BoundedBufferSem
{
    private static final int BUFFER_SIZE = 5;
    private int count; // number of items in the buffer
    private int in; // points to the next free position in the buffer
    private int out; // points to the next full position in the buffer
    private Object[] buffer;
    Semaphore mutex, full, empty;
    public BoundedBuffer()
    {
        // buffer is initially empty
        count = 0;
        in = 0;
        out = 0;
        buffer = new Object[BUFFER_SIZE];
        mutex = //A → new Semaphore(1);
        full = //B → new Semaphore(0);
        empty = //C → new Semaphore(BUFFER_SIZE);
    }

    //producer calls this method
    public void insert(Object item) {
        try {
            //D → empty.acquire();
            //E → mutex.acquire();
        }
        catch (InterruptedException e) {
        }
        buffer[in] = item;
        in = (in + 1) % BUFFER_SIZE;
        count++;
        if (count == BUFFER_SIZE)
            System.out.println("Producer Entered " + item +
                               " Buffer FULL");
        else
            System.out.println("Producer Entered " + item +
                               " Buffer Size = " + count);
        //F → mutex.release();
        //G → full.release();
    }
}
```

```
41 // consumer calls this method
42 public Object remove() {
43     Object item;
44     try {
45         //H → full.acquire();
46         //I → mutex.acquire();
47     }
48     catch (InterruptedException e) {
49     }
50     item = buffer[out];
51     out = (out + 1) % BUFFER_SIZE;
52     count--;
53     if (count == 0)
54         System.out.println("Consumer Consumed " + item +
55                             " Buffer EMPTY");
56     else
57         System.out.println("Consumer Consumed " + item +
58                             " Buffer Size = " + count);
59     //J → mutex.release();
60     //K → empty.release();
61     return item;
62 }
63 }
```

3. From the BoundedBuffer class in the picture. The Semaphore is used to provide Synchronization. Based on the algorithm we learned in class.

The code at //A is ____

Hide Results

43/43 Students Answered

A	mutex.acquire();	7%
B	mutex.release();	0%
C	new Semaphore(1);	86%
D	new Semaphore(0);	7%
E	new Semaphore(BUFFER_SIZE);	0%

4. From the BoundedBuffer class in the picture. The Semaphore is used to provide Synchronization. Based on the algorithm we learned in class.

The code at //B is ____

Hide Results

43/43 Students Answered

A	mutex.acquire();	0%
B	mutex.release();	2%
C	new Semaphore(1);	5%
D	new Semaphore(0);	77%
E	new Semaphore(BUFFER_SIZE);	7%

4.10.2024 (Deadline)

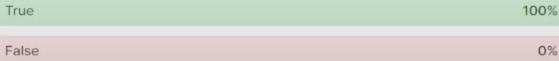
< Back to Results Table

< 1 of 17 >

1. Deadlock involves 2 or more processes.

Hide Results

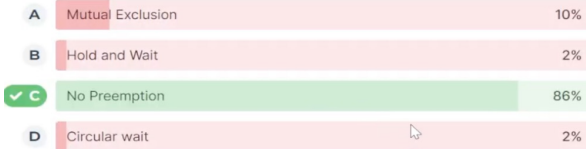
42/43 Students Answered



3. A resource can be released only voluntarily by the process holding it.

Hide Results

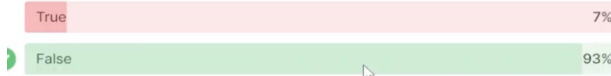
42/43 Students Answered



5. If only one condition in the 4 conditions for deadlock is true then deal lock occurs.

Hide Results

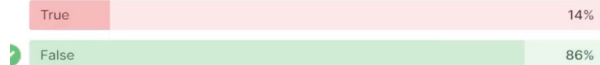
42/43 Students Answered



7. Request edge of a resource-allocation graph has a direction from a resource to a process. *True - Assi Edge*

Hide Results

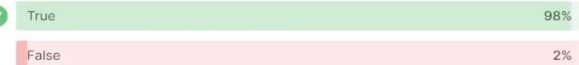
42/43 Students Answered



9. Usually, the operating systems we use today do not handle deadlock for users' processes.

Hide Results

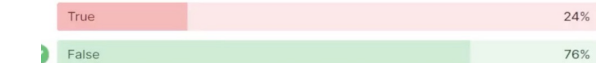
42/43 Students Answered



11. Giving each resource a number and requiring a process to request resources in increasing order can prevent a no-preemption condition.

Hide Results

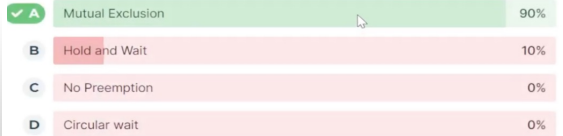
42/43 Students Answered



2. Only one process at a time can use a resource.

Hide Results

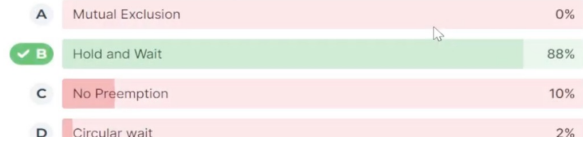
42/43 Students Answered



4. A process holding at least one resource and waiting for resources held by other processes.

Hide Results

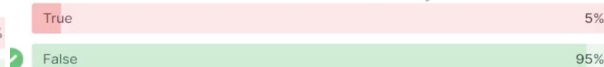
42/43 Students Answered



6. Resource-Allocation Graph to model deadlock is an undirected graph.

Hide Results

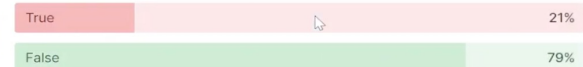
42/43 Students Answered



8. If a cycle is detected in a resource-allocation graph then deadlock.

Hide Results

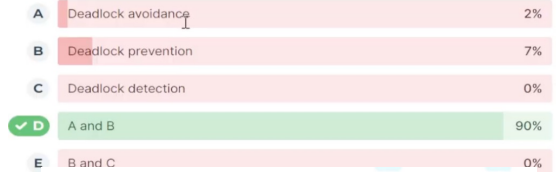
42/43 Students Answered



10. Deadlock handling mechanism(s) that do(es) not allow a deadlock to occur. *Protection → allow → then solve*

Hide Results

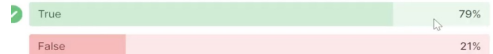
42/43 Students Answered



12. Using a resource-allocation graph to avoid deadlock can work if there is only one resource for each type of resource.

Hide Results

42/43 Students Answered



13. One condition for deadlock avoidance is that a process must inform an OS of all resources it needs to use to finish its task before it starts.

Hide Results

42/43 Students Answered

True	90%
False	10%

15. OS will change a claim edge to a(an) _____ to test for the safe state.

Hide Results

42/43 Students Answered

☒ A	Assignment Edge	33%
☐ B	Claim Edge	10%
☐ C	Request Edge	57%
☐ D	Resource Edge	0%

17. From C program in the picture It is a deadlock free program.

Hide Results

42/43 Students Answered

☐	True	21%
☒	False	79%

No Explanation

4*10 (Deadlock)

14. An edge in a resource-allocation graph for deadlock avoidance with a direction from a process to a resource to tell an OS that it may request for the resource is called _____.

Hide Results

42/43 Students Answered

☐ A	Assignment Edge	10%
☒ B	Claim Edge	79%
☐ C	Request Edge	12%
☐ D	Resource Edge	0%

16. The system is safe if the cycle is not detected after changing the claim age to the assignment edge.

Hide Results

42/43 Students Answered

☒	True	76%
☐	False	24%

```

1  #include <pthread.h>
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <unistd.h>
5  #include <semaphore.h>
6  sem_t mutex1;
7  sem_t mutex2;
8  void *func1 (void *param);
9  void *func2(void *param);
10 int main() {
11     int i = 0;
12     pthread_t tid1, tid2;
13     pthread_attr_t attr;
14     pthread_attr_init(&attr);
15     sem_init(&mutex1, 0, 1);
16     sem_init(&mutex2, 0, 1);
17     pthread_create(&tid1, &attr, func1, NULL);
18     pthread_create(&tid2, &attr, func2, NULL);
19     pthread_join(tid1, NULL);
20     pthread_join(tid2, NULL);
21     sem_destroy(&mutex1);
22     sem_destroy(&mutex2);
23 }
24 void *func1 (void *param) {
25     sem_wait(&mutex1);
26     sem_wait(&mutex2);
27     printf("Hello\n");
28     sem_post(&mutex1);
29     sem_post(&mutex2);
30     pthread_exit(0);
31 }
32 void *func2 (void *param) {
33     sem_wait(&mutex2);
34     sem_wait(&mutex1);
35     printf("Hello\n");
36     sem_post(&mutex2);
37     sem_post(&mutex1);
38     pthread_exit(0);
39 }

```

Real Memory Management (11-10-24)

Load & Compile

1. The dynamic address binding is done at the _____ time.

Hide Results 43/43 Students Answered

A	compile	0%
B	load	2%
✓ C	run	81%
D	A and B	5%
E	B and C	12%

3. The memory address accessed by the memory unit.

Hide Results 43/43 Students Answered

A	Logical address	2%
✓ B	Physical Address	95%
C	Virtual Address	0%
D	A or B	2%
E	A or C	0%

5. .dll file of MS Windows is an example of _____.

Hide Results 43/43 Students Answered

A	Memory Management Unit	2%
✓ B	Dynamic Linking	44%
C	Dynamic Loading	0%
D	A and B	0%
E	B and C	53%

7. For the dynamic linking concept, the loaded library can be shared among processes.

Hide Results 43/43 Students Answered

✓ True	100%
False	0%

9. For fixed partitions scheme, the maximum number of processes that can be loaded into memory at a time is the number of partitions.

Hide Results 43/43 Students Answered

✓ True	95%
False	5%

11. Assume that the base register stores 1000 and limit stores 2000, the request to address 2100 is considered legal.

Hide Results 43/43 Students Answered

True	21%
False	79%

No Explanation

2. The memory address generated by the CPU is called _____.

Hide Results 43/43 Students Answered

A	Logical address	23%
B	Physical Address	0%
C	Virtual Address	2%
D	A or B	0%
✓ E	A or C	74%

4. Translating logical address to physical address is the task of _____

Hide Results 43/43 Students Answered

✓ A	Memory Management Unit	93%
B	Dynamic Linking	2%
C	Dynamic Loading	2%
D	A and B	2%
E	B and C	0%

6. The main concept of dynamic linking is to load only the required parts of the program first, the libraries will be loaded when they are needed.

Hide Results 43/43 Students Answered

✓ True	91%
False	9%

8. Dividing memory into sections, each section can have different size, but after the system is started we cannot change the number of sections and their size is called _____.

Hide Results 43/43 Students Answered

A	Dynamic Loading	0%
✓ B	Fixed Partitions	98%
C	Variable Partitions	2%
D	A or B	0%

10. For fixed partitions scheme, base and limit registers are used to prevent a process to access the partition that belongs to others. The limit register is used to store the beginning of the partition where the base register is used to store the size of the partition.

Hide Results 43/43 Students Answered

True	2%
False	98%

12. Assume that the base register stores 1000 and limit stores 2000, the request to address 300 is considered legal.

Hide Results 43/43 Students Answered

True	12%
✓ False	88%

12. Assume that the base register stores 1000 and limit stores 2000, the request to address 900 is considered legal.

Hide Results

43/43 Students Answered

True	12%
False	88%

14. One problem of fixed partitions scheme is _____.

Hide Results

43/43 Students Answered

internal fragmentation	100%
external fragmentation	0%
worst-fit	0%
best-fit	0%

16. To solve the problem of variable partitions by combining the adjacent free blocks into one large block is called _____.

Hide Results

43/43 Students Answered

Compaction	16%
Coalescing	84%
Best-Fit	0%
First-Fit	0%
Worst-Fit	0%

18. The compaction cannot be done if address binding is done at runtime.
cannot be done at compiling / load time.

Hide Results

43/43 Students Answered

True	60%
False	40%

20. If a memory has 4 free partitions from the lower to higher addresses as follows, a = 16 GB, b = 5 GB, c = 14 GB, d = 30 GB. Assume that a new process enters the system and request 12 GB. What partition that the process will be allocated if the Best-Fit algorithm is used? (the closest size...)

Hide Results

43/43 Students Answered

a	0%
b	2%
c	98%
d	0%
not enough memory to allocate.	0%

13. Assume that the base register stores 1000 and limit stores 2000, the request to address 3000 is considered legal.

Hide Results

43/43 Students Answered

True	7%
False	93%

15. Variable partitions scheme can cause the internal fragmentation.

Hide Results

43/43 Students Answered

True	7%
False	93%

17. To solve the problem of variable partitions by rearranging the processes that are distributed in the memory to make a large contiguous block is called _____.

Hide Results

43/43 Students Answered

Compaction	91%
Coalescing	9%
Best-Fit	0%
First-Fit	0%
Worst-Fit	0%

19. If a memory has 4 free partitions from the lower to higher addresses as follows, a = 16 GB, b = 5 GB, c = 14 GB, d = 30 GB. Assume that a new process enters the system and request 12 GB. What partition that the process will be allocated if the First-Fit algorithm is used?

first check → can fit → use it

Hide Results

43/43 Students Answered

a	70%
b	2%
c	26%
d	2%
not enough memory to allocate.	0%

21. If a memory has 4 free partitions from the lower to higher addresses as follows, a = 16 GB, b = 5 GB, c = 14 GB, d = 30 GB. Assume that a new process enters the system and request 12 GB. What partition that the process will be allocated if the Worst-Fit algorithm is used?

(largest one)

Hide Results

43/43 Students Answered

a	2%
b	0%
c	2%
d	95%
not enough memory to allocate.	0%

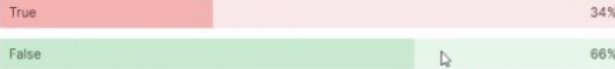
Real Memory Management (11-10-24)

Real Memory Management (11-10-24) — Non-Contiguous Scheme

1. For paging scheme, the physical memory will be divided into fixed-sized blocks called page. frame - fixed size block

Hide Results

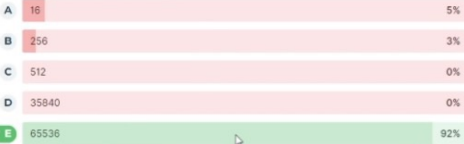
38/43 Students Answered



3. If a CPU uses 16-bit address, the maximum memory that a CPU can address is _____ bytes. 2^{16}

Hide Results

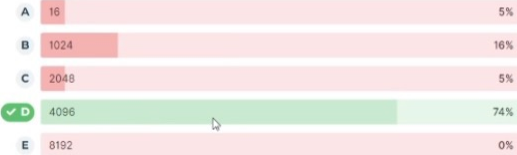
38/43 Students Answered



5. If a CPU uses 16-bit address if a page size is 16 bytes, how many maximum pages in a program? $16 - 4 = 12; 2^{12}$

Hide Results

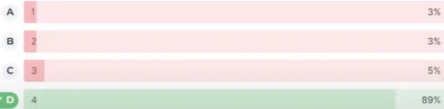
38/43 Students Answered



7. From the picture, how many pages are there in the program?

Show Results

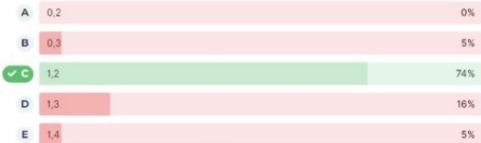
38/43 Students Answered



9. From the picture, the logical address for "g" is page _____ and offset _____

Hide Results

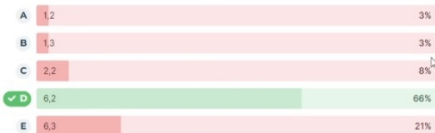
38/43 Students Answered



10. From the picture, "g" is stored in frame _____ and offset _____.

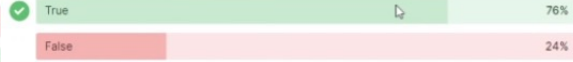
Hide Results

38/43 Students Answered



Hide Results

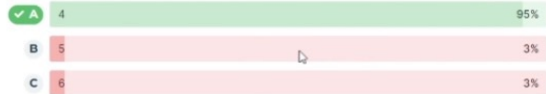
38/43 Students Answered



4. If a CPU uses 16-bit address if a page size is 16 bytes, how many bits (from the 16 bits) are used for page offset? $4 \text{ bits} = 2^4$

Hide Results

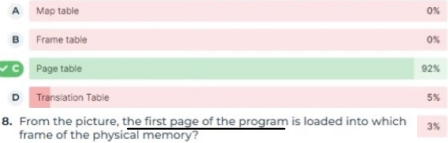
38/43 Students Answered



6. For paging scheme, a _____ is used for logical address to physical address translation.

Hide Results

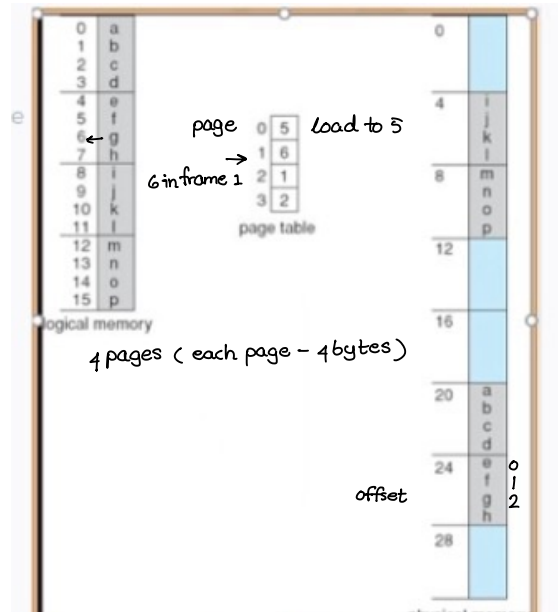
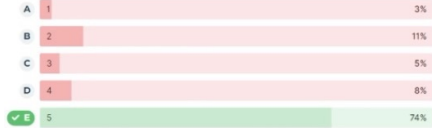
38/43 Students Answered



8. From the picture, the first page of the program is loaded into which frame of the physical memory?

Hide Results

38/43 Students Answered



Real Memory Management (11-10-24) — Non-Contiguous Scheme)

< Reports

Real Memory Management (Noncontiguous Scheme) (1-2024)

Friday, October 11, 2024 at 3:06 PM

< Back to Results Table

11. Assume that TLB access time is 0.43 microsecond. Memory access time is 1 microsecond. Effective Access Time (EAT) is 1.35 microsecond. What is the percent of Hit Ratio?

Hide Results 38/43 Students Answered

A	50	8%
B	60	0%
C	70	22%
D	80	3%
E	90	68%

13. Assume that the vi program requires 5 pages for its code and 2 pages for its data. If the shared-page concept is applied, how many frames will be shared among processes of the vi program?

Hide Results 38/43 Students Answered

A	1	8%
B	2	11%
C	3	8%
D	4	5%
E	5	68%

15. For segmentation scheme, segment table is used for logical address to physical address translation.

Hide Results 38/43 Students Answered

True	100%
False	0%

17. In a segment table, the field that is used to specify the size of the segment is called ____.

Hide Results 38/43 Students Answered

A	base	0%
B	initial	3%
C	limit	95%
D	start	3%
E	end	0%

19. From the following segment table; limit base 1000 1400 400 6300 400 4300 1100 3200 1000 4700 If a logical address in the format segment-number: offset is 2:400. The physical address is ____

Hide Results 38/43 Students Answered

A	1700	0%
B	6600	3%
C	4600	16%
D	3500	0%
E	error	82%

$$EAT = 2 + \epsilon - \alpha$$

$$\alpha = 2 + \epsilon - EAT$$

$$2 + 0.25 - 1.35$$

Formula !

12. For paging scheme, a page table has a(n) ____ bit to do the memory protection.

Hide Results 38/43 Students Answered

A	dirty bit	0%
B	no use bit	0%
C	wrong bit	5%
D	valid-invalid bit	89%
E	table entry	5%

14. The main idea of the segmentation scheme is to divide a program into parts that may not have the same size. However those parts must be loaded into the adjacent parts of the memory.

Hide Results 38/43 Students Answered

True	18%
False	82%

16. In a segment table, the field that is used to specify the start address of the segment is called ____.

Hide Results 38/43 Students Answered

A	base	95%
B	initial	0%
C	limit	5%
D	start	0%
E	end	0%

18. From the following segment table; limit base 1000 1400 400 6300 400 4300 1100 3200 1000 4700 If a logical address in the format segment-number: offset is 1:300. The physical address is ____

Hide Results 38/43 Students Answered

A	1700	8%
B	6600	87%
C	4600	3%
D	3500	0%
E	error	3%

20. For segmentation scheme, the external fragmentation is still exist.

Hide Results 38/43 Students Answered

True	92%
False	8%

Virtual Memory (18.10.24)

1. For virtual memory concept, logical address space can be larger than physical address space.

Hide Results

38/43 Students Answered

True 92%

False 8%

3. For demand paging virtual memory concept, the page table is still used for logical address to physical address translation.

Hide Results

38/43 Students Answered

True 89%

False 11%

5. Reference to a page that is not in the memory will cause _____.

Hide Results

38/43 Students Answered

A page not found 3%

B page error 0%

✓ C page fault 95%

D page exception 3%

E page format 0%

7. If a page fault occurs and there is no free frame available, OS will terminate the program.

Hide Results

38/43 Students Answered

True 11%

✓ False 89%

9. If the memory has 3 frames, and now page 2, 0, and 1 are loaded into these frames respectively. Assume the page reference string of the remaining program is as follow 7 0 1 2 0 3 0 4 2 3 0. Also, assume that the FIFO page replacement is used. If page 7 is referenced, the page that will be replaced is _____.

Hide Results

38/43 Students Answered

A 0 3%

B 1 0%

✓ C 2 97%

D 3 0%

E 4 0%

2. For virtual memory concept, the entire program does not need to be loaded at the same time.

Hide Results

38/43 Students Answered

✓ True 95%

False 5%

4. A bit in the page table that is used to determine that a page is in a frame or not is _____.

Hide Results

38/43 Students Answered

A flagged bit 3%

B loaded bit 0%

C checked bit 5%

✓ D valid-invalid bit 92%

E page fault bit 0%

6. If the thing in the previous question occurs, an OS will do the followings except _____.

Hide Results

38/43 Students Answered

A Get empty frame 3%

B Swap page into frame 0%

C update page table 16%

D restart the instruction that caused the page fault 0%

✓ E change the value of valid-invalid bit to invalid 82%

8. In general, if we add more memory to the system the page fault rate will decrease.

Hide Results

38/43 Students Answered

✓ True 87%

False 13%

10. If the memory has 3 frames, and now page 2, 0, and 1 are loaded into these frames respectively. Assume the page reference string of the remaining program is as follow 7 0 1 2 0 3 0 4 2 3 0. Also, assume that the FIFO page replacement is used. How many page faults will occur when the process starts executing the remaining parts of the program.

Hide Results

38/43 Students Answered

A 3 0%

B 4 3%

C 5 3%

D 6 32%

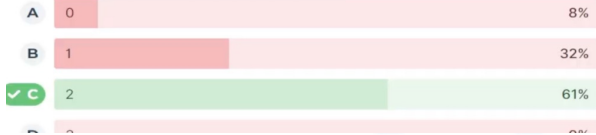
✓ E 7 63%

Virtual Memory (18.10.24)

11. If the memory has 3 frames, and now page 2,0, and 1 are loaded into these frames respectively. Assume the page reference string of the remaining program is as follow 7 0 1 2 0 3 0 4 2 3 0. Also, assume that the Optimal page replacement is used. If page 7 is referenced, the page that will be replaced is 1.

Hide Results

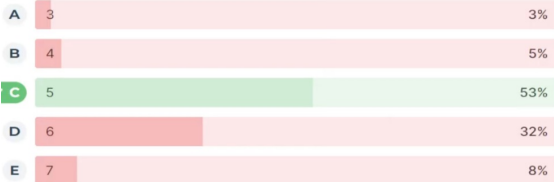
38/43 Students Answered



12. If the memory has 3 frames, and now page 2,0, and 1 are loaded into these frames respectively. Assume the page reference string of the remaining program is as follow 7 0 1 2 0 3 0 4 2 3 0. Also, assume that the Optimal page replacement is used. How many page faults will occur when the process starts executing the remaining parts of the program.

Hide Results

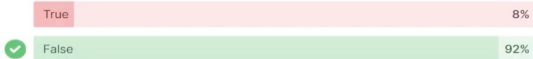
38/43 Students Answered



13. Optimal page replacement is the algorithm that most OSes use since it gives the minimum page fault rate.

Hide Results

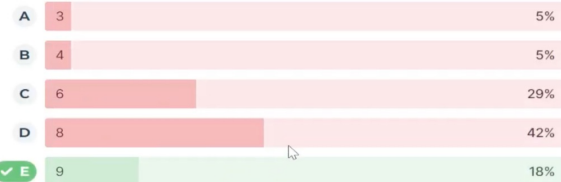
38/43 Students Answered



14. Assume that the memory has 3 frames and page reference string is as follows; 7 0 1 2 0 3 0 4 2 3 0. The page replacement algorithm is LRU. How many page faults will occur when this program is executed?

Hide Results

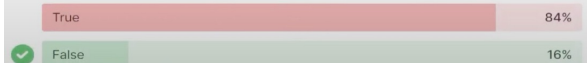
38/43 Students Answered



15. The frames that will be selected as replacement frames for a process will be chosen from only the frames that OS allocated to the process.

Hide Results

38/43 Students Answered



Virtual Memory (18.10.24)

16. If a system is busy swapping pages in and out, it can be said that _____ occurred.

Hide Results

38/43 Students Answered

A	busy waiting	0%
B	page fault	0%
C	dispatching	5%
✓ D	thrashing	95%
E	page replacement	0%

18. Assume that working set window is 10. At the time t1, assume that the process p1 has the following page reference string 2 5 1 5 7 7 7 5 1. Assume that the process p2 has the following page reference string 1 2 0 5 4 5 6 6 7. How many minimum numbers of free frames must be available to avoid thrashing?

{1,2,5,7}
{0,1,2,3,4,5,6,7}

Hide Results

38/43 Students Answered

A	7	8%
B	10	8%
✓ C	12	82%
D	14	3%

Assume that one row of an array requires the space of 1 page. How many page faults will occur for the array initialization as shown in the code?

Ans

Hide Results

38/43 Students Answered

A	10	11%
B	128	0%
C	256	5%
✓ D	2560	79%
E	16384	5%

20.

```
public class Main {  
    public static void main(String[] args) {  
        int [][] a = new int[10][256];  
        for(int i = 0; i < 10; i++) {  
            for(int j = 0; j < 256; j++) {  
                a[i][j] = 0;  
            }  
        }  
    }  
}
```

Ans

Assume that one row of an array requires the space of 1 page. How many page faults will occur for the array initialization as shown in the code?

17. If an OS cannot detect the situation in the previous question, it may load more processes into the system since the CPU is not well utilized.

Hide Results

38/43 Students Answered

✓ True	92%
False	8%

19. From the following Java program

```
public class Main {  
    public static void main(String[] args) {  
        int [][] a = new int[10][256];  
        for(int j = 0; j < 256; j++) {  
            for(int i = 0; i < 10; i++) {  
                a[i][j] = 0;  
            }  
        }  
    }  
}
```

Assume that one row of an array requires the space of 1 page. How many page faults will occur for the array initialization as shown in the code?

Assume that one row of an array requires the space of 1 page. How many page faults will occur for the array initialization as shown in the code?

Hide Results

38/43 Students Answered

✓ A	10	82%
B	128	5%
C	256	5%
D	2560	8%
E	16384	0%